ASIANZU BENJAMIN CHRIS

Health Data Analyst • Machine Learning • Public Health Informatics

+256 780 165 621 • b3ntana@gmail.com • Kampala, Uganda (Open to Remote)

chris-asianzu.github.io/healthcare.html • github.com/chris-asianzu • linkedin.com/in/chris-asianzu-9b52a538b

PROFESSIONAL SUMMARY

Health Data Analyst and Machine Learning practitioner with demonstrated experience delivering predictive analytics and digital health systems in real-world public health settings across East Africa. Built and deployed a Community Fever & Malaria Surveillance System — live Streamlit dashboard tracking disease clusters across 4 Ugandan districts. Developed a disease outbreak risk prediction system for a regional public health NGO achieving 84% model accuracy and estimated 28% reduction in malaria cases. Co-developed a live health records system at Maracha Hospital, West Nile Uganda. Open to remote roles with WHO, UNICEF, international NGOs, and global health analytics organizations.

CORE COMPETENCIES

Machine Learning: scikit-learn, random forest, gradient boosting, logistic regression, ROC-AUC, recall optimization

Health Informatics: disease surveillance, outbreak detection, health records systems, rural health data infrastructure, KoboCollect/KoboToolbox

Data Engineering: ETL pipelines, KoboToolbox API, multi-source data integration, lag variable engineering, SQLAlchemy, SQLite

Languages & Tools: Python (pandas, numpy, scikit-learn, matplotlib), R (ggplot2, tidyverse), SQL (PostgreSQL, MySQL)

Visualization: Power BI (DAX, heatmaps, risk dashboards), Streamlit, folium maps, Tableau, ggplot2, matplotlib

Data Sources: NASA climate datasets, World Bank demographic indicators, district health surveillance reports, KoboCollect mobile surveys

Soft Skills: Stakeholder reporting, executive dashboards, curriculum design, NDA client delivery, field team coordination

PROFESSIONAL EXPERIENCE

Data Analyst & IT Officer | Muni Labs, West Nile, Uganda

2023 – Present

- Built and deployed a Community Fever & Malaria Surveillance System — mobile data collection via KoboCollect, Python ETL pipeline via KoboToolbox API, SQLite database, and a live Streamlit dashboard with interactive district heatmaps and weekly fever/malaria trend analysis across 4 Ugandan districts.
- Co-developed and deployed a Health Records System at St. Joseph's Hospital, Maracha — live digital infrastructure replacing paper-based patient records in a rural healthcare setting.
- Conducted infrastructure gap analysis and issued technical recommendations (server specs, LAN architecture) enabling real-time multi-user system deployment.
- Designed and delivered a 12-week Introduction to Data Analysis course for the Graduate Studies Academy — 5 cohorts completed, training university students and working professionals across agriculture, health, and education sectors.
- Provided ongoing data analysis, IT support, and software solutions across NGO health and research operations.

Freelance Health Data Analyst | Regional Public Health NGO, East Africa (NDA)

2024

- Contracted to build a predictive analytics system for early disease outbreak detection across 45 districts in East Africa.
- Integrated 7 years of multi-source surveillance data (180,000+ observations) from district health offices, NASA climate datasets, and World Bank demographic indicators.
- Engineered lag features linking rainfall patterns to malaria cases 4–6 weeks later. Correlations: rainfall vs malaria 0.62, temperature anomaly vs outbreak intensity 0.48.
- Final model: Random Forest — 84% accuracy, ROC-AUC 0.89, recall 0.81.
- Delivered Power BI dashboard with national risk heatmap, 30-day outbreak probability forecasting, and intervention prioritization panel.
- Simulation models estimated 28% potential reduction in malaria cases through early intervention.

Course Instructor – Python & Programming | ReedInc Org, West Nile, Uganda

Feb 2024

- Developed and delivered beginner Python programming curriculum for community learners.

Biochemistry Teacher | Secondary School, Uganda

2023 – Present

- Teach O-level and A-level Biology and Chemistry aligned to Uganda National Curriculum.

KEY PROJECTS

Community Fever & Malaria Surveillance System

Live Deployment — Built with Abiyo Bruce | community-fever-surveillance.streamlit.app

- End-to-end digital surveillance: KoboCollect mobile form → Python ETL via KoboToolbox API → SQLite database → live Streamlit dashboard with interactive heatmaps and weekly trend analysis.
- 1,000 records across 4 Ugandan districts (Arua, Gulu, Lira, Kampala). Successfully deployed and collecting real field data.
- Key findings: fever spikes 4–6 weeks after high rainfall; low mosquito net usage correlates with higher malaria positivity.
- Tools: Python, KoboCollect, KoboToolbox API, SQLite, Streamlit, folium, pandas

Disease Outbreak Risk Prediction System

Paid NDA Client Project

- ML pipeline: integrated 180,000+ observations over 7 years from district health offices, NASA climate data, and World Bank demographics.
- Random Forest: 84% accuracy, ROC-AUC 0.89, recall 0.81. Estimated 28% reduction in malaria cases through early intervention.
- Delivered Power BI risk dashboard with 30-day forecasting and intervention prioritization panel.
- Tools: Python, pandas, scikit-learn, Random Forest, SQL, Power BI, NASA API, World Bank Data

Maracha Hospital Health Records System

Live Deployment — Muni Labs | St. Joseph's Hospital, West Nile Uganda

- Co-developed digital health records infrastructure replacing paper-based patient records with a centralized LAN-connected database.
- System demonstrated September 2024; full deployment pending hospital hardware infrastructure upgrade.
- Conducted gap analysis and issued server/LAN architecture recommendations for production rollout.

E-Commerce Revenue Intelligence System

Commercial Analytics | github.com/chris-asianzu/ecommerce-revenue-intelligence

- 98,666 orders, 3 years of data. Found 87.7% churn rate with R'1.38M recoverable revenue. Full pipeline: Python → PostgreSQL → Power BI → board presentation.
- Tools: Python, PostgreSQL, Power BI, pandas, SQLAlchemy

Employee Attrition Analysis

R Statistical Project | github.com/chris-asianzu/Employee-attrition-analysis

- Statistical analysis of employee attrition: hypothesis testing, logistic regression model, ggplot2 visualizations revealing key turnover drivers.
- Tools: R, ggplot2, tidyverse, Logistic Regression, Hypothesis Testing

EDUCATION

Bachelor of Science in Education (Biology & Chemistry) | Muni University, Arua, Uganda — Feb 2023

Uganda Advanced Certificate of Education | Our Lady of Africa SS, Mukono — Dec 2017

CERTIFICATIONS

- Microsoft Certified: Data Analyst Associate (Power BI)
- Certified Python Programmer — DataCamp, 2024
- Certified R Programmer — R Foundation, 2022
- Data Analysis with Python — freeCodeCamp
- Career Skills in Data Analytics — LinkedIn Learning